FLOWSTATE / RESEARCH REPORT

Synthetic BMTC Boardings

Evidence review, calibration priors, and v1 configuration recommendations

Prepared	12 October 2026
Geography	Bengaluru metropolitan area, Karnataka, India
Use	Research and configuration decision; no implementation assumed
Primary question	What can defensibly calibrate synthetic BMTC passenger events when no public stop-by-stop boarding series is available?
Output affected	`flowstate.bus.scheduled_stop.v1`: the synthetic fields `boardings`, `alightings` and `onboard_leaving` on each scheduled stop call

Core finding

Use Bengaluru-specific published bus demand shape and trip-length evidence, but keep scale, stop allocation, and physical occupancy explicitly synthetic until BMTC route, time and passenger records can be accessed.

What this report is not

It is not an estimate of observed BMTC ridership. The unofficial GTFS feed supplies schedules and stop locations only. Every passenger quantity FlowState generates is an assumption-driven synthetic value and must be labelled as one.

Figures in this report are carried from the FlowState research memo dated 12 October 2026. Evidence classes follow that memo's labels. Reference numbers [R1]–[R11] are listed at the end.

How to read this report

The report has two parts so that the reasoning and the numbers do not get tangled.

Part I is the argument in plain prose: what the synthetic stream can honestly claim, what it borrows from published Bengaluru evidence, and what it refuses to claim. It deliberately carries no raw figures, so a reader can follow the logic without being anchored on a number.

Part II is the reference: the recommended configuration, every figure with its source and evidence class, the charts, the keep/adopt/reject table, the evidence gaps, and the references. Implementers should work from Part II. Part I explains why each line there reads the way it does.

Reading the evidence marks

- Filled marker: official statistic or peer-reviewed study, with period, geography and definition stated.
- Hollow marker: press restatement or a model choice. Useful context or a working assumption, not a measurement.

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PART I

Decisions and reasoning

The argument in prose. No raw figures; see Part II for every number.

I.1 The question and the short answer

FlowState emits one record for every scheduled BMTC stop call. The timetable says where and when a bus calls. It says nothing about how many people are aboard. So every passenger field on those records is manufactured, and the real question is how much of that manufacture can lean on published Bengaluru evidence rather than on convenient guesses.

The short answer is a split. Two things can be borrowed from published local work with reasonable honesty: the *shape* of demand across the day, and the *distribution of trip lengths*. Three things cannot: the true *scale* of a weekday, the *allocation of boardings to individual stops*, and the *physical occupancy* of a bus. Those three stay synthetic, are recorded as assumptions, and are tested as ranges rather than defended as facts.

I.2 What the numbers are, and what they are not

A boarding is a passenger boarding event. It is the closest match to what BMTC reports as passengers carried, and it is the right target when summing generated records. It is not a person: the same individual can board several times in a day, and a single door-to-door journey with a transfer produces more than one boarding. Counting people, journeys, passenger-kilometres and vehicle-kilometres are different quantities, and none of them can be recovered from boardings alone.

The same discipline applies to the schedule. A scheduled call is a possible event, not an observed one. Departure counts, stop-time counts and route counts cannot be promoted into boarding weights and then described as demand. The trip-conservation and non-negative-load checks in the generator are useful mechanical invariants, but passing them says nothing about whether the generated demand resembles BMTC's.

I.3 How big should the synthetic city be?

The published figures do not form one series. They come from different periods, different datasets and different definitions: a planning-document table, an audit statistic, a one-week sample of electronic ticket records that leaves out pass holders, and a recent press restatement of a ministerial statement. They cannot be joined with a smooth growth curve, and the memo explicitly advises against trying.

The practical consequence is that the weekday total is a *model choice*. The recommendation is to place a provisional centre between the audited pre-pandemic figure and the current headline, run a lower and a higher sensitivity case around it, and record the target year and policy regime beside the number. A strict replay of the last audited pre-pandemic year uses the audited figure instead. Because the alignment window falls after the free-travel scheme for women began, the pre-scheme figure is not automatically the right scale for it. Day-type multipliers must also be checked so that calendar effects are not counted twice.

I.4 Seats, standing room and load factors

Two easily confused ideas live here. The seating figure that has a source is an average number of *seats* per bus from a dated state survey. It is not total capacity, because it excludes standing passengers. If the configuration parameter is meant to hold total capacity, the sourced number is semantically wrong for it, and that has to be settled before the numeric value is touched.

The load-factor figures from the audit are built from ticket revenue. They describe how well a service earns against its capacity, not how many people were aboard at an instant, so they cannot be inverted into an occupancy cap. The earlier suggestion of a particular multiplier traced back to a social-media claim about passenger counts and road-transport rules; that does not establish a statutory limit and must not be described as one.

The multiplier may remain as an *engineering guardrail* that stops the generator producing impossible loads, provided it is configurable, labelled synthetic, and sensitivity-tested at lower values. A better model later

maps each trip to a vehicle variant with approved seat and standee capacities.

I.5 Does the metro take riders from the buses?

The tempting move is to shrink bus demand by some factor because metro ridership has grown. The evidence does not support one. The strongest local study found that buses and the metro mostly serve different markets: buses carry the shorter trips, the metro the longer ones, even on some routes that run alongside it. Most bus routes overlap the metro on a map, but a small minority carry meaningful numbers of passengers travelling parallel to it. Spatial overlap is not demand overlap.

That study also predates the Yellow Line, and no matched route-level before-and-after record exists to estimate what the new line changed. A record metro day cannot show how many bus boardings were lost, because metro growth can come from new trips, transfers, diverted private-vehicle trips and feeder journeys that *add* to bus use. A survey in which metro adopters report previously using the bus describes those adopters, not the whole bus network. The citywide substitution multiplier therefore stays neutral. That is the conservative choice of not applying an unsupported adjustment, not a claim that the metro has no effect.

I.6 What can be borrowed: time of day and trip length

A Bengaluru bus curve built from ticket issue times shows a clear morning and evening peak, with a large share of daily demand concentrated in the two combined windows around them. It also shows a sharp fall in the evening that the authors tie partly to the end of scheduled service, not only to passenger preference. This is a better shape constraint than inferring bus demand from metro exits and ride-hail trips, and it should become the preferred external check.

It has limits. The ticket records omit pass holders, the observations are old, and the paper publishes a curve and summary rather than a free complete hourly table. The memo's instruction is firm: preserve provenance if the figure is digitised, version the extraction, and do not invent twenty-four hourly weights from a summary. Until then, the existing metro and ride-hail blend stays as a fallback, with a new check that the generated peak windows are consistent with the published share.

Trip length is the second borrowable quantity. Two historical citywide averages agree closely, and a distribution by distance band is published. These work as *soft targets* wherever route distances can be computed from ordered stops or paths. They do not support inferring how many stops a trip has, and the pandemic-year distribution and the metro-area subset must not be mistaken for normal citywide travel. Where route geometry is missing or malformed, uncertainty is preserved rather than papered over with a precise origin–destination matrix.

I.7 What stays synthetic

Stop attraction has no observed distribution behind it, because the feed defines candidate calls and nothing more. The convention that early stops board and late stops alight is therefore a generator design choice, not a behavioural finding, and it must be checked for artefacts. Alighting placement follows the same rule: trip-length bands steer it softly, and nothing pretends to reproduce a real origin–destination pattern.

The feed itself is unofficial. Its maintainers warn that some routes may be missing and some timings wrong. Generated values must never be labelled as automatic passenger counts, the provenance of every field must separate real schedule rows from synthetic passenger fields, and the generated taxi files that FlowState's own audit rejected cannot be used to tune bus demand under any circumstances.

I.8 What would change the picture

Nearly every open question in this report closes with one kind of access: BMTC's own records. Ticketing data by route, trip and hour including pass validations; a fleet master with approved seat and standee capacities; passenger counts keyed to route, direction, stop and time; and a post-Yellow-Line route comparison that controls for the free-travel scheme, fares, service supply and route changes. None of these is public today. The distinction matters: electronic ticketing data *exists* inside BMTC, and the limitation is public availability, not absence.

Until such records can be reached, the stream is a calibrated synthetic event generator. Uncertain priors are carried as ranges and scenario runs, and every number keeps its evidence label attached.

PART II

Reference: configuration, evidence and sources

Every figure with its period, definition and evidence class.

II.1 Recommended near-term configuration

Table 1. Recommended v1 configuration.

Field / behaviour	Recommendation	Evidence status	Decision
`assumed_weekday_boardings`	4.6m provisional centre; test 4.0m and 4.8m. Use 3.31m for a strict FY2019-20 scenario.	Official historical baseline plus recent press-reported scale; the centre is a model choice.	Adopt provisionally; record target year and regime.
`assumed_bus_capacity`	41.3 seats only if the parameter means seats; no sourced total standee-inclusive capacity.	State survey through Nov 2018; no current type-level mapping.	Adopt with semantic caveat; prefer type approval later.
`assumed_max_load_factor`	1.4 may remain only as a synthetic guardrail; sensitivity 1.0 / 1.2 / 1.4.	No local measured peak occupancy supports one exact factor.	Keep as assumption, not as fact.
Hour profile	Peaks 08–10 and 17–19; 47–51% of demand in the 08–11 plus 17–20 blocks.	Peer-reviewed study; Jan 2020 and Mar 2021 ETM samples.	Update evidence note; current blend only as fallback.
`metro_substitution_multiplier`	1.0 citywide.	No causal citywide coefficient for the Yellow Line period.	Keep v1 default.
Day-type multipliers	Keep v1 until daily bus data by day type is available.	No open current calibration series.	Keep v1, mark unvalidated.
Stop attraction / boarding	No observed stop-demand distribution; GTFS defines candidate calls only.	No open route or stop count series in FlowState.	Keep explicitly synthetic.
Alighting / trip distribution	Use published trip-length bins as soft targets where route distances are available.	Historical fare-derived estimate, not a current OD matrix.	Adopt as soft target only.

II.2 Daily scale

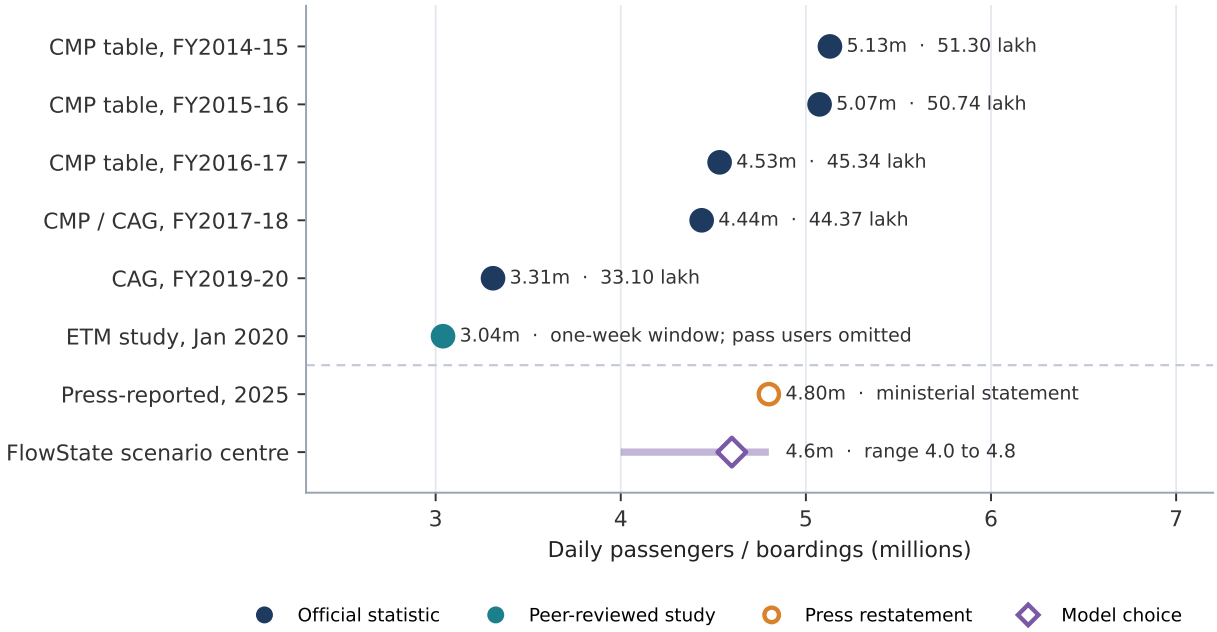


Figure 1. Reported daily BMTC scale by source. Points are plotted by source, not by date, and are deliberately not joined: the periods, datasets and definitions differ. The dashed line separates the historical series from the current-scale reference and the FlowState scenario. ○ marks press restatements and model choices.

Table 2. Observed and reported scale anchors.

Period	Value	Definition / source context	Evidence class	Use
2014-15	51.30 lakh/day	BMTC passengers carried per day, CMP table	Official report citing BMTC	Historical context
2015-16	50.74 lakh/day	Same CMP series	Official report citing BMTC	Historical context
2016-17	45.34 lakh/day	Same CMP series	Official report citing BMTC	Historical context; secondary series differ
2017-18	44.37 lakh/day	CMP and CAG report this value	Official report / CAG statistic	Historical anchor
2019-20	33.10 lakh/day	CAG BMTC passenger base	Official audit statistic	Best audited pre-pandemic anchor for this FY
Jan 2020	3.04m/day	Average daily ridership in a one-week ETM analysis window	Peer-reviewed study output	Separate study estimate; incomplete pass capture
2025 (press)	48 lakh/day	Ministerial statement reported for BMTC	Press restatement of official statement	Contemporary scale check, not an audited weekday mean

Recommended prior. The FlowState alignment window is 26 October 2024 to 15 April 2026, so the pre-scheme FY2019-20 count is not automatically the right scale. For a first 2025/26 scenario use 4.6m as the provisional centre and test 4.0m and 4.8m. This is a model choice within the reported current scale, not a measured weekday average. When replaying FY2019-20, use 3.31m. Store target regime and year, and make sure day-type multipliers do not double-count calendar effects [R1, R5, R6].

`assumed_weekday_boardings` should mean the expected sum of generated boardings across stop calls for the simulated service day, subject to the implementation's transfer and through-rider semantics. Do not convert it to passenger-kilometres or treat it as unique persons.

II.3 Fleet, seating, capacity and load factor

What is actually sourced

- CAG reports 6,799 buses in the BMTC fleet on 31 March 2022 (ordinary and premium) and 5,547 schedules. This is an audit-period count, not a current fleet mix by type [R1].
- Karnataka Economic Survey 2018-19 reports an average seating capacity of 41.3 and 45.00 lakh daily passengers through 30 November 2018. The passenger figure is partial-year; 41.3 is seats, not total occupancy [R2].
- AIS-052 (Rev.1) says seating and standee capacities should appear in the vehicle body type-approval certificate. Generic micro/mini/midi/standard categories are not a verified crosswalk to BMTC ordinary, Vayu Vajra, Vajra, feeder or specific EV variants [R8].

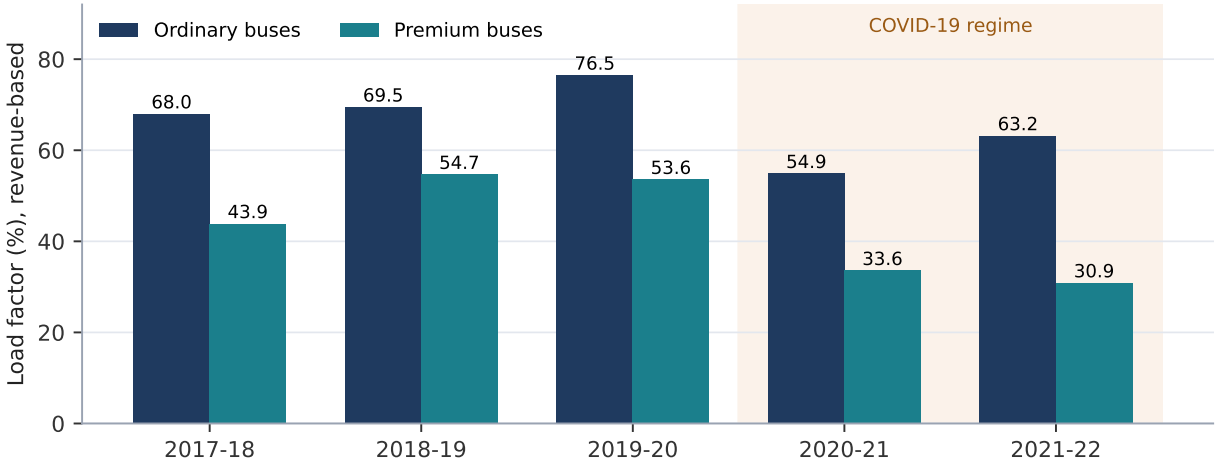


Figure 2. CAG load factors by financial year. These are calculated using ticket revenue and are historical operational indicators, not observed peak onboard headcounts. 2020-22 is a separate demand regime [R1].

The 1.4 guardrail

The suggested 1.4 multiplier was tied to a social-media claim about 65–70 passengers and road-transport rules. That does not establish a universal statutory limit. If v1 uses it to prevent impossible synthetic loads it may remain a configurable engineering guardrail, clearly labelled synthetic, and sensitivity-tested at 1.0, 1.2 and 1.4.

Table 3. Illustrative cap on `onboard_leaving` if `assumed_bus_capacity` means seats. Derived arithmetic (41.3 × factor), not a sourced capacity.

Load-factor setting	1.0	1.2	1.4 (v1 guardrail)
Implied cap (passengers)	41.3	49.6	57.8

If `assumed_bus_capacity` means seats, 41.3 (about 41) is a dated fleet-average prior. If it means total standing-plus-seated capacity, 41.3 is not semantically correct. Resolve the meaning before changing the value. Never describe tested load-factor values as measured BMTC occupancy.

II.4 Modal share: context, not a multiplier

- The Bengaluru Comprehensive Mobility Plan uses a 2016 household interview survey. Its all-trip summaries put public transport at roughly 32%, two-wheelers about 27%, walking about 26%, auto-rickshaws

about 7% and cars/taxis/shared taxis about 7%. The public-transport category combines bus, minibus, school bus, chartered bus and metro; it is not a bus-only share [R3].

- A 2025 metro/feeder integration study reports that 65.6% of respondents had used a regular city bus before adopting metro. This is a prior-mode share among surveyed metro adopters, not the fraction of all bus passengers who switched, and not a causal estimate of net bus ridership lost [R11].

Decision: no current standalone bus/metro/auto split suitable for scaling generated boardings was found. Keep the 2016 survey as context only; do not use it to manufacture route-level counts.

II.5 Metro opening and bus demand

Gadepalli et al. (2022) compared the metro with the BMTC network, service and ticketing data for Phase-1 conditions, with demand observations for January 2020 and March 2021 [R4]. Their findings:

- 1,903 of 2,203 routes (86%) had some spatial overlap with the metro, but overlap alone did not imply parallel passenger travel.
- Only 734 of those routes had passenger trips with both ends within 400 m of metro stations.
- 132 routes (6% of total) had at least 10% of demand travelling at least 2 km parallel to the metro; 27 routes had at least 25% of demand in that category.
- From January 2017 to January 2020 the operational fleet grew 12% and citywide effective kilometres grew 6%; effective-km growth in the metro influence area was about 2%. Demand there was described as marginally lower, presumably related to the metro, but no citywide substitution coefficient was established.

These results predate the Yellow Line, which opened in August 2025 (11 August; station count 69 to 83 in the RTI data). FlowState has a structural change in metro observations at that date but no matched BMTC route-level panel to estimate diversion.

Decision: keep `metro_substitution_multiplier` $= 1.0$ citywide. If route-level analysis is ever done, estimate effects only on routes with matched OD evidence and a credible comparison [R4, R7].

II.6 Ticketing, APC, and route/stop counts

- The 2022 study used ETM backups with route-wise origin, destination, ticket issue time and fare. It captured about 1.7 million ticketed users per day pre-COVID but omitted student and employee pass users, estimated at around half of total bus demand. Bus–bus and bus–metro transfers were unavailable, so one journey could appear as several legs [R4].
- No public downloadable BMTC dataset or API with stop-by-hour observations, route-by-hour boardings, APC counts or complete ticket/pass validations was identified [R9].
- The accepted source is the unofficial Vonter/bmtc-gtfs feed (feed date 2026-09-07; about 9,960 stop records in the project inventory). Project notes warn that some routes may be missing and some timings inaccurate [R9, R10].

Decision: treat GTFS as possible stop calls and a spatial/schedule scaffold only. Do not set boarding weights from stop-time or route counts and call them observed demand. Do not label generated boardings or alightings as APC.

II.7 Trip length and alighting placement

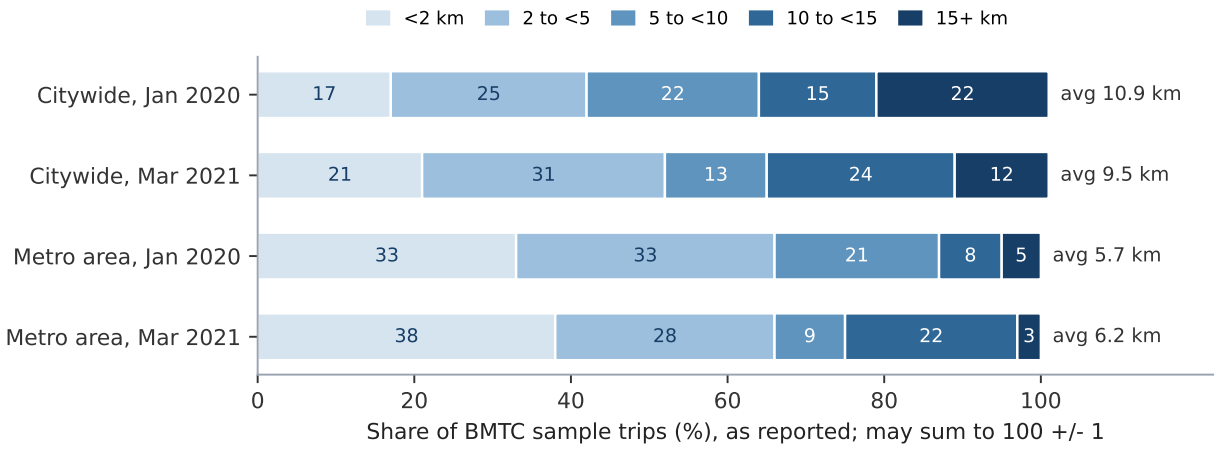


Figure 3. BMTC sample trip-length distributions from fare-derived distances [R4]. Percentages are as published and may not total 100 because of rounding.

Table 4. Average bus trip length cross-checks.

Source / sample	Average	Note
CMP, citing RITES study 2016 [R3]	10.7 km	Citywide historical cross-check
Gadepalli et al., citywide Jan 2020 [R4]	10.9 km	Fare-derived
Gadepalli et al., citywide Mar 2021 [R4]	9.5 km	Pandemic-affected
Metro influence area, Jan 2020 / Mar 2021 [R4]	5.7 / 6.2 km	Subset; do not apply citywide

Decision: use 10.7–10.9 km as a historical citywide cross-check and the bins as a soft constraint where route distances can be computed from ordered stops or paths. Do not infer stops per trip from kilometres without geometry or inter-stop distances. If route paths are missing or malformed, preserve uncertainty instead of inventing a precise OD matrix.

II.8 Hour-of-day demand prior

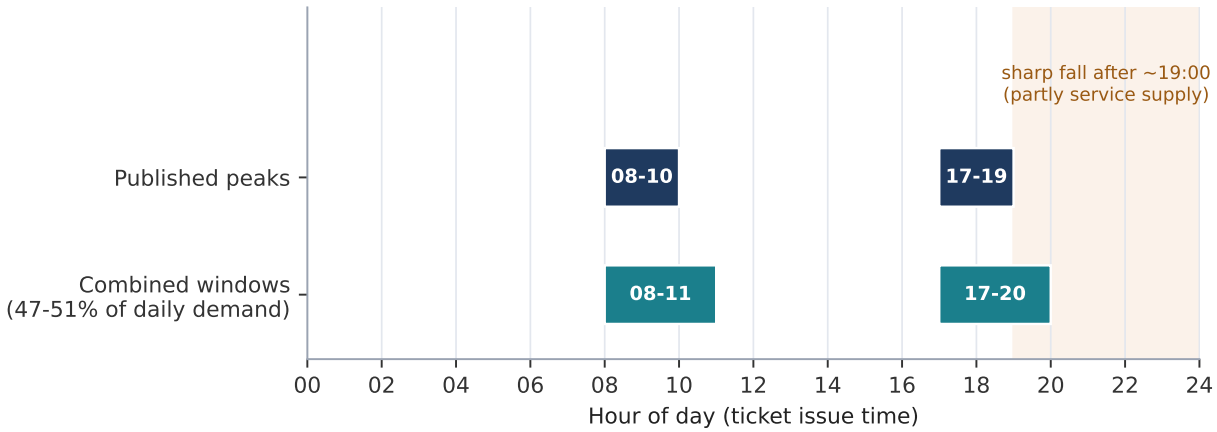


Figure 4. Published peak and combined-peak windows [R4]. Only the windows are shown; no hourly curve has been digitised and none should be inferred from this chart.

Table 5. Published hourly indicators (ETM ticket issue times, Jan 2020 and Mar 2021) [R4].

Indicator	Published result
Morning peak	08:00–10:00
Evening peak	17:00–19:00
Combined peak windows	08:00–11:00 and 17:00–20:00
Share of daily demand in combined windows	47–51%
Mar 2021 scheduled services terminating by 18:00	1,236 of 5,438 (23%)

The paper ties the sharp fall in demand after about 19:00 partly to scheduled service availability. This curve is better than shaping bus demand from metro exits and ride-hail alone, but it is not a full raw citywide dataset: ETM tickets exclude pass users, the observations are old, and only a curve and summary are published.

Decision: document the BMTC-specific curve as the preferred external shape constraint. If exact 24-hour weights can be reliably digitised, preserve provenance and version the extraction. Until then retain the existing metro/ride-hail blend as fallback and add checks that the combined peak windows are consistent with the 47–51% range. Do not invent all 24 hourly percentages from the summary.

II.9 Keep, adopt, reject

Table 6. Candidate treatments and decisions.

Candidate	Decision	Reason
4.6m weekday baseline for 2025/26	Adopt provisionally, not as a published statistic	Current press scale is 4.8m daily; no weekday-only series. Test 4.0m and 4.8m.
3.31m/day in FY2019-20	Adopt for the historical FY2019-20 scenario	CAG-reported passenger base; not the current policy regime.
41.3 average seats	Adopt only if the parameter means seats	Dated fleet-average seat figure; not total passenger capacity.
1.4 as official legal / observed load factor	Reject	Social-media claim is insufficient; CAG load factor is revenue-based, not peak occupancy.
2016 32% public-transport share as bus share	Reject	CMP category combines several public-transport modes.
65.6% prior bus users among metro adopters as substitution	Reject	Conditional metro-adopter survey, not a citywide net change.
Metro ridership growth reduces bus boardings	Reject	No matched causal citywide estimate; modes can substitute, complement or generate trips.
GTFS route/stop counts as passenger demand	Reject	Schedule is not passenger observation.
Early stops board, late stops alight as local behaviour	Reject as an empirical claim	No BMTC stop-level evidence; if retained as a generator, label it synthetic and check for artefacts.
Generated taxi files rejected by FlowState	Reject completely	Project source audit identifies generated or mislabelled files; they cannot tune bus demand.

II.10 Evidence gaps and how to close them

- **Current baseline by day type:** daily passenger totals or validated ticket/pass data for a representative 8–12 week period, with weekday, Saturday, Sunday, holiday, school-day and major-event flags.
- **Route-time demand:** de-identified ETM ticket issue records by route, trip and hour, including pass validations; document whether “trip” means a ticketed leg or a complete journey.
- **Vehicle capacity mapping:** a fleet master keyed by registration or variant with seat and standee capacities from type-approval certificates.

- **Stop-level counts / APC:** passenger counts or manual load surveys keyed by route, direction, stop, time and trip ID, ideally including the maximum-load point.
- **Updated household travel survey:** current mode share and trip distances with separate bus, metro, auto and taxi/ride-hail categories and a stated geography.
- **Post-Yellow-Line route comparison:** comparable BMTC route-level counts before and after 11 August 2025, controlling for Shakti, fares, service supply, work and education patterns, and route changes.

Until these exist, treat generated stop values as synthetic calibrated events, not observations. Use ranges and scenario runs for uncertain priors and keep the evidence label attached to every configuration value.

II.11 Evidence and citation rules for future updates

- **Official statistic:** government audit, operator report, official survey or report, or technical standard. Cite the period, geography and definition.
- **Survey estimate / empirical study:** state the sample, year, target population and known bias. A result among metro adopters is not a citywide mode share.
- **Press restatement of an official statement:** useful for a current headline figure but weaker than a published daily series with definitions. Mark it as such.
- **Model choice / missing evidence:** mark as an assumption. An engineering guardrail is not measured just because it is numerically precise.

For FlowState changes, cite the source in the relevant ADR or configuration explanation, record the retrieval date and source file or hash, and keep separate fields for target year and regime, central value, sensitivity range and evidence class. Do not silently rebase history as new reports arrive.

> **Bottom line**
>
> The best-supported improvement over v1 is to add Bengaluru-specific bus demand evidence for hourly shape and trip lengths, while keeping scale, stop-level allocation and physical occupancy explicitly synthetic until BMTC route, time and passenger records are available. Do not use GTFS as ridership, turn revenue-based load factors into physical occupancy, or deduct bus demand simply because metro ridership rises.

References

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[R3] DULT, Government of Karnataka, with partner agencies. *Comprehensive Mobility Plan for Bengaluru* (2020). Household interview survey 2016; bus trip length attributed to the RITES Study, 2016. https://dult.karnataka.gov.in/assets/front/pdf/Comprehensive_Mobility_Plan.pdf

[R4] Gadepalli, R., Gumireddy, S., Bhamidipati, S., & Cats, O. (2022). Impact of introducing a metro line on urban bus services. *Case Studies on Transport Policy*, 10(2), 940–947. DOI 10.1016/j.cstp.2022.03.007. https://repository.tudelft.nl/file/File_4c6065de-6639-45eb-9340-75c69b12956e

[R5] The Times of India. "Transport minister slams calls to end BMTC monopoly, terms move anti-public." October 2025. Ministerial statement of about 4.8 million BMTC daily riders; press restatement. https://timesofindia.indiatimes.com/city/bengaluru/transport-minister-slams-calls-to-end-bengaluru-metropolitan-transport-corporation-monopoly-terms-move-anti-public/articleshow/124560415.cms

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[R10] Vonter/bmtc-gtfs, unofficial BMTC GTFS source. Feed date 2026-09-07; schedule-only role. https://github.com/Vonter/bmtc-gtfs

[R11] "Evaluating multimodal integration of metro and feeder bus services in Bengaluru using integrated analytical and perception based methods." *Multimodal Transportation* (2025), DOI 10.1016/j.multra.2025.100287. Prior-mode result is conditional on metro-service respondents. https://doi.org/10.1016/j.multra.2025.100287